LESSON PLAN · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

# Lesson 6: Project: Jumpy Platformer

Module 2 | 50 minutes | Project: platformer | Grades 3–5 · ages 9–11

**Objective:** Plan, build and test a mini platformer with gravity, jumping, walking, a collectable and a goal.

**Success criteria**

- My hero walks and jumps with gravity.
- Collectables change the score.
- My level has a goal or hazard and was playtested.

## Key vocabulary

- **platformer**: A game where you run and jump between places.
- **collectable**: Something the player picks up for points.
- **hazard**: Something the player must avoid.

## You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

## Timed lesson plan (50 min)

| Time | Stage | What happens |
|---|---|---|
| 5 min | **Warm-up (unplugged)** | Level sketch (5 min): draw a level on paper with a start, three coins, a hazard and a goal. |
| 5 min | **Watch & plan** | Play the project video; learners plan on the project brief. |
| 25 min | **Build** | Build-along steps, then Which sprite should have this script?. Check in halfway: everyone has something working. |
| 7 min | **Test & peer feedback** | Swap and test each other's work: two stars and a wish. |
| 8 min | **Improve & submit** | Improve one thing; Project: Jumpy Platformer. |

## Unplugged warm-up

Level sketch (5 min): draw a level on paper with a start, three coins, a hazard and a goal.

## Teaching the idea

The platformer integrates velocity-based physics, input polling, per-sprite collision rules and win/lose conditions. Emphasise decomposition: each sprite owns its behaviour.

**Analogy:** A play: every actor (sprite) knows their own lines (scripts).

**Live demo:** Play a finished mini level; then open the scripts sprite by sprite.

## Questions to ask

| Ask | Listen for |
|---|---|
| Which sprite owns the gravity script? | The Hero. |
| How does the coin know it was collected? | It checks touching Hero. |
| How does the game end? | Reaching the goal or a score target. |

## Differentiation

**Support**
Provide the hero physics; learners add a coin and a goal.

**Stretch**
Multiple levels with backdrops (switch backdrop when the goal is reached).

## Common misconceptions

| Misconception | How to fix it |
|---|---|
| The hero script should check every coin | Each coin checks the hero itself. |
| Gravity needs to be in every sprite | Only sprites that fall need it. |

## Assessment

Project rubric (also on the project brief):

| Criterion | 1 · Beginning | 2 · Developing | 3 · Secure | 4 · Excellent |
|---|---|---|---|---|
| **Physics** | No gravity | Falls at a constant speed | Velocity gravity, floor and jump | Tuned feel, no mid-air jumps, screen edges |
| **Controls** | Hard to control | Steps only | Smooth walk and jump | Polished controls |
| **Game rules** | Missing | Coins only | Coins and a hazard or goal | Coins, hazard and a clear win |
| **Design** | Unplayable | Playable | Fun and fair | Creative level with a challenge curve |
| **Testing & reflection** | Not tested | Tested alone | Playtested; tuning explained | Several improvements explained |

**Exit ticket**
One number you tuned to make the game feel better.

**Homework**
Draw your next level and write which new sprite it needs.

## Answer key — Kitabash activities

**1. Which sprite should have this script?** (perform · categorize)

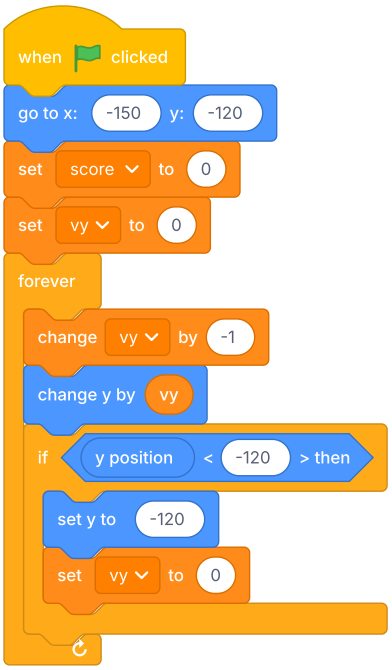
**Hero:** Gravity loop, Jump on space, Walk with arrow keys
**Coin:** If touching Hero → change score
**Hazard:** If touching Hero → send the hero back

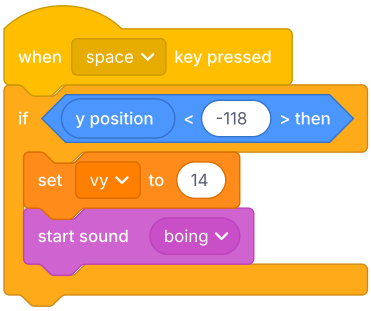
## 2. Project: Jumpy Platformer (probe · code lab)

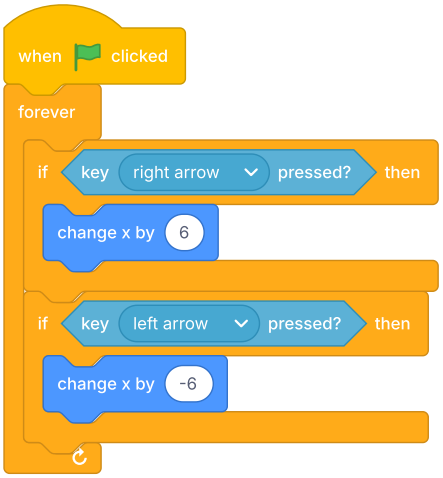
**Goals checked:** Walks · Jumps · Gravity · Coins score · Score starts at 0

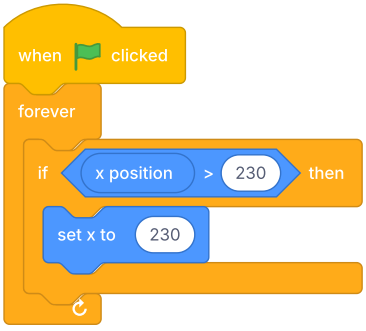
**Example project:**

**Hero**

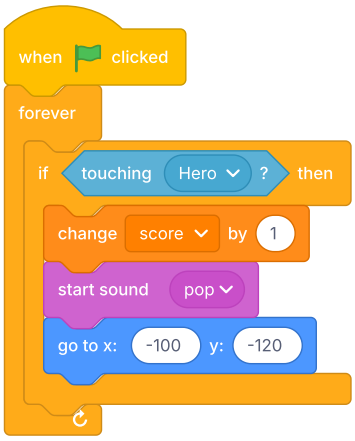

when clicked
go to x: -150 y: -120
set score to 0
set vy to 0
forever
change vy by -1
change y by vy
if y position < -120 > then
set y to -120
set vy to 0


when space key pressed
if y position < -118 > then
set vy to 14
start sound boing


when clicked
forever
if key right arrow pressed? then
change x by 6
if key left arrow pressed? then
change x by -6


when clicked
forever
if x position > 230 then
set x to 230

## Coin




## Answer key — worksheet

1. **Key words:** platformer = A game where you run and jump between places. · collectable = Something the player picks up for points. · hazard = Something the player must avoid.
2. **Which sprite should have this script?:** Hero: Gravity loop, Jump on space, Walk with arrow keys · Coin: If touching Hero → change score · Hazard: If touching Hero → send the hero back
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.